

INFO510 Bayesian Modelling and Inference –

Guidelines for final project

Fall 2025

The goals are twofold:

- a) to utilize what you have learned from the course;
- b) an opportunity to work on your own data.

Dataset

Choosing a dataset is something you should do carefully but also relatively quickly. Choose a dataset from any online repository e.g. Kaggle, TidyTuesday etc., or from any research paper that you might have read recently, or you can use your own data.

Questions

You have to think of a **minimum of 2 questions** that you can address using Bayesian methods for data analysis. When selecting the questions and subsequent methods for analysing it, pay attention to the type of data you have e.g. binary, categorical, continuous, etc.

Proposal writing and submission – 2 page max

The proposal should contain the following;

1. Dataset selected
2. Project goals and a **minimum of 2 questions** based on the dataset selected
3. Tentative plan of analysis e.g. what prior you will use, model use e.g. Bayesian regression model along with the MCMC sampling, methods for model validation and prediction
(you can use stan (R) or JAGS (R) for analysis, specifically for setting the model and MCMC sampling)
4. Expected outcomes (what results you think you will get after the analysis)

Keep in mind the proposal is a rough idea of what analysis you will perform; the plans can change down the road when you actually start analyzing the data. However, please put in some effort to write a detailed proposal as it will help you get started.

PS: If possible, avoid changing the question completely or picking up a different question after proposal is submitted.

Report Write-up and Submission

(*Do not write a lengthy project report, keep it concise)**

The project report should contain the following parts.

1. **Introduction (1-2 paragraphs):** Brief introduction to the dataset and the questions (detailed description/introduction of the questions will come in their respective sections). Write it imagining that your project is a standalone document, and the grader has no prior knowledge of the dataset.

2. Question 1

- a. **Introduction (1-2 paragraphs):** Introduction to the question and what parts of the dataset are necessary to answer the question. Also discuss why you're interested in this question.
- b. **Methods (1 page max):** Describe what types of methods you used to address your question.
Include the following contents –
 - What are the priors used and the logic behind using the priors
 - What is the model used for analysis, along with the MCMC sampling parameters
 - Check convergence with diagnostics like trace plots and Gelman-Rubin statistics
 - Generate credible intervals for model coefficients
 - Evaluate the predictive performance using posterior predictive checks
 - Model validation
- c. **Data Analysis and results:** In this section, provide the code* that generates the results for your analysis. Write the results obtained, show plots, figures and tables wherever necessary
*The code has to be submitted as .rmd or .r file
- d. **Discussion (1-3 paragraphs):** In this section, interpret the results obtained
- e. **Reference section (optional)**

3. Question 2

Same structure outlined for Question 1, but for your new question

Note - The project report should be submitted as a PDF document, any codes generated should be submitted as .RMD, .R or .ipynb (if you are using python). It doesn't matter if you are using R or Python, but the codes should be properly annotated and easy to follow, and reproducible.

Grading (Total project points = 100 pts)

Proposal - 15 pts

Report writing - 35 pts

Introduction – 3 pts

Question 1:

Introduction – 2 pts

Methods – 5 pts

Data Analysis and result – 6 pts

Discussion – 3 pts

Question 2:

Introduction – 2 pts

Methods – 5 pts

Data Analysis and result – 6 pts

Discussion – 3 pts

Presentation – 50 points

Deadlines –

Proposal submission – **October 30, 2025 at 11.59 PM**

Final Project report submission – **November 28, 2025 at 11:59 PM**

Project presentations – **December 1, 2025 onwards (depending on # of groups)**